# LEVEL UP ACADEMY

Class 10 | GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Drill: 40 Objective Questions (GSEB Section A practice)

*Chapter / Topic: Mixed chapters · Week 9 of the 90-day plan*

*For teacher use only. Do not distribute to students.*

### CHOOSE THE CORRECT OPTION

**Q1.** Which of the following is a decomposition reaction?
**Answer:** (c) $2H_2O \rightarrow 2H_2 + O_2$

**Q2.** Which of these turns red litmus blue?
**Answer:** (b) Soap solution (basic)

**Q3.** The chemical formula of washing soda is:
**Answer:** (d) $Na_2CO_3 \cdot 10H_2O$

**Q4.** Which metal is the best conductor of electricity?
**Answer:** (c) Silver

**Q5.** Which of these is a saturated hydrocarbon?
**Answer:** (b) $C_3H_8$ (propane)

**Q6.** Bile is produced by the:
**Answer:** (a) liver

**Q7.** The opening and closing of stomata is controlled by the:
**Answer:** (d) guard cells

**Q8.** Which hormone is secreted by the testes?
**Answer:** (c) Testosterone

**Q9.** Pollen grains are produced in the:
**Answer:** (d) anther

**Q10.** A trait that is expressed only when both copies of the gene are the same is called:
**Answer:** (a) recessive

**Q11.** The SI unit of refractive index is:
**Answer:** (b) it has no unit (it is a ratio of two speeds)

**Q12.** The image formed on the retina of the eye is:
**Answer:** (d) real and inverted

**Q13.** 1 volt is equal to:
**Answer:** (a) 1 J/C (V = W/Q)

**Q14.** The direction of the force on a current-carrying conductor in a magnetic field is given by:
**Answer:** (a) Fleming's left-hand rule

**Q15.** The ozone layer absorbs harmful:
**Answer:** (b) ultraviolet radiation

### FILL IN THE BLANKS

**Q16.** When a substance gains oxygen during a reaction, it is said to be _____.
**Answer:** oxidised

**Q17.** Packets of chips are flushed with _____ gas to prevent rancidity.
**Answer:** nitrogen

**Q18.** A solution that turns phenolphthalein pink has a pH _____ than 7.
**Answer:** greater (more)

**Q19.** An alloy of copper and zinc is called _____.
**Answer:** brass

**Q20.** The general formula of alkanes is _____.
**Answer:** $C_nH_{2n+2}$

**Q21.** Blood vessels that carry blood towards the heart are called _____.
**Answer:** veins

**Q22.** The gap between two neurons is called a _____.
**Answer:** synapse

**Q23.** In Mendel's monohybrid cross, the $F_2$ phenotypic ratio is _____.
**Answer:** 3 : 1

**Q24.** The power of a convex lens of focal length 50 cm is _____.
**Answer:** +2 D

**Q25.** Only about _____% of the energy at one trophic level is passed to the next.
**Answer:** 10

## TRUE OR FALSE

**Q26.** All decomposition reactions need heat energy.
**Answer:** False. Some need light or electricity.

**Q27.** Aluminium oxide is an amphoteric oxide.
**Answer:** True

**Q28.** Graphite is a good conductor of electricity.
**Answer:** True

**Q29.** Haemoglobin mainly carries carbon dioxide in the blood.
**Answer:** False. Haemoglobin mainly carries oxygen.

**Q30.** The cerebellum maintains posture and balance.
**Answer:** True

**Q31.** The resistance of a wire increases when its area of cross-section increases.
**Answer:** False. Resistance decreases as the area increases.

**Q32.** A ray passing through the centre of curvature of a concave mirror retraces its path after reflection.
**Answer:** True

**Q33.** CFCs help to build up the ozone layer.
**Answer:** False. CFCs deplete the ozone layer.

## ANSWER IN ONE WORD OR PHRASE

**Q34.** Name the type of reaction in which a more reactive metal pushes a less reactive metal out of its salt solution.
**Answer:** Displacement reaction

**Q35.** Name the salt used to make plaster casts for fractured bones.
**Answer:** Plaster of Paris

**Q36.** Name the green coating that forms on copper vessels left in moist air.

**Answer:** Basic copper carbonate

**Q37.** Name the hormone that controls the blood sugar level.

**Answer:** Insulin

**Q38.** Name the defect of vision that develops with old age.

**Answer:** Presbyopia

**Q39.** Name the rule that gives the direction of the magnetic field around a straight current-carrying wire.

**Answer:** Right-hand thumb rule

**Q40.** Name the organisms that break down the dead remains of plants and animals.

**Answer:** Decomposers (bacteria and fungi)